

8.6 Several factors, interactions and covariates

With one factor, rank deficiency is a nuisance, and a coding removes it once and for all. With two or more factors, or a factor together with a numeric regressor, it becomes informative. The rank of the model matrix now depends on which combinations of levels were observed. Whether a comparison is estimable depends on how the design links the levels involved. This section works out the three cases that occur most often: two factors without interaction, two factors with interaction, and a factor with a numeric covariate.

8.6.1 Two factors without interaction

Let factor A have levels $i = 1, \dots, a$ and factor B levels $j = 1, \dots, b$. Call a pair (i, j) a **cell**. A cell is *occupied* if at least one observation has that combination, and we assume every level of each factor occurs in some occupied cell. The *additive* model is

$$E(y) = \mu + \alpha_i + \beta_j \quad \text{for an observation in cell } (i, j), \quad (8.6.1)$$

with model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B]$ built from the two indicator matrices. The number of parameters is $p = 1 + a + b$. Since $\mathbf{Z}_A \mathbf{1}_a = \mathbf{Z}_B \mathbf{1}_b = \mathbf{1}$, the vectors $(1, -\mathbf{1}_a^\top, \mathbf{0}^\top)^\top$ and $(1, \mathbf{0}^\top, -\mathbf{1}_b^\top)^\top$ always lie in $\mathcal{N}(\mathbf{X})$, so $\text{rank}(\mathbf{X}) \leq a + b - 1$. Whether equality holds, and which differences $\alpha_i - \alpha_k$ are estimable, depends on the pattern of occupied cells.

Picture the design as a graph. The vertices are the a levels of A and the b levels of B , and each occupied cell (i, j) is an edge joining level i of A to level j of B . The design is **connected** if this graph is connected, that is, if one can walk from any level to any other through occupied cells.

Theorem 8.6.1. Connectedness and estimability in the additive model

In the additive model [Eq. 8.6.1](#), let the design graph have c connected components. Then:

- (a) $\text{rank}(\mathbf{X}) = a + b - c$;
- (b) $\alpha_i - \alpha_k$ is estimable iff levels i and k of A lie in the same component, and likewise for $\beta_j - \beta_l$;
- (c) if the design is connected, the estimable functions are exactly the linear functions of the cell means $\mu + \alpha_i + \beta_j$ over *all* cells, occupied or not. In particular every difference between levels of the same factor is estimable.

Proof. A vector $(m, \mathbf{u}^\top, \mathbf{v}^\top)^\top$ lies in $\mathcal{N}(\mathbf{X})$ iff $m + u_i + v_j = 0$ for every occupied cell (i, j) . Fix such a vector. If (i, j) and (k, j) are both occupied, then $u_i = u_k$. If (i, j) and (i, l) are both occupied, then $v_j = v_l$. Following edges, u is constant on the A -levels of each component, say $u_i = t_C$ for all A -levels i in component C , and then $v_j = -m - t_C$ for the B -levels of C . Conversely, any choice of m and of numbers $t_1, \dots, t_c$, one for each component, defines in this way a vector satisfying all the equations. So $\mathcal{N}(\mathbf{X})$ is parameterized linearly and injectively by $(m, t_1, \dots, t_c)$, and it has dimension $c + 1$. Rank-nullity gives $\text{rank}(\mathbf{X}) = 1 + a + b - (c + 1)$, which is (a).

For (b), by Theorem 8.3.1(c), $\alpha_i - \alpha_k$ is estimable iff $u_i - u_k = 0$ for every null vector, that is, iff $t_{C(i)} = t_{C(k)}$ for all choices of the t 's. This holds iff $C(i) = C(k)$. The argument for B

is the same.

For (c), if $c = 1$, then (b) makes every $\alpha_i - \alpha_k$ and $\beta_j - \beta_l$ estimable. Pick an occupied cell (i_0, j_0) . Then $\mu + \alpha_i + \beta_j = (\mu + \alpha_{i_0} + \beta_{j_0}) + (\alpha_i - \alpha_{i_0}) + (\beta_j - \beta_{j_0})$ is estimable for every (i, j) . Conversely, every estimable function is a combination of the means of occupied cells (Theorem 8.2.2), and these are among the cell means. ■

Part (c) contains a point worth stating separately. In a connected additive design, the mean of an *empty* cell is estimable. The data never observed that combination, but additivity lets it be pieced together from cells that were observed: $\mu + \alpha_i + \beta_j$ is the mean of an occupied cell (i, j') , plus the difference $\beta_j - \beta_{j'}$ learned elsewhere. The estimate is only as good as the additivity assumption, which cannot be checked at the empty cell. This is extrapolation of a subtle kind.

Example. Two islands

Take $a = 4$, $b = 5$ and seven occupied cells: $(1, 1), (1, 2), (2, 2), (2, 3)$ and $(3, 4), (4, 4), (4, 5)$ (levels numbered from 1). The design graph has two components: rows 1–2 with columns 1–3, and rows 3–4 with columns 4–5. By Theorem 8.6.1, $\text{rank}(\mathbf{X}) = 4 + 5 - 2 = 7$, one less than $a + b - 1 = 8$. The difference $\alpha_1 - \alpha_2$ is estimable, since rows 1 and 2 are linked through column 2, but $\alpha_1 - \alpha_3$ is not. The two islands could sit at any vertical offset from each other without changing any fitted value. Occupying a single further cell, $(3, 3)$, joins the islands. The rank rises to 8, and $\alpha_1 - \alpha_3$ becomes estimable. The listing checks each statement with the rank test Theorem 8.3.1(h). The script also checks the rank formula on 200 random designs.

```
import numpy as np

def additive_matrix(cells, a, b):
    """Model matrix [1, row indicators, column indicators] for occupied (i, j) cells
    ."""
    rows = np.array([i for i, j in cells])
    cols = np.array([j for i, j in cells])
    return np.column_stack([np.ones(len(cells)), np.eye(a)[rows], np.eye(b)[cols]])

def estimable(lam, X):
    """lambda is estimable iff appending it to the rows of X does not raise the rank
    ."""
    return np.linalg.matrix_rank(np.vstack([X, lam])) == np.linalg.matrix_rank(X)

a, b = 4, 5
cells = [(0, 0), (0, 1), (1, 1), (1, 2), (2, 3), (3, 3), (3, 4)] # two separate
                                blocks
X = additive_matrix(cells, a, b)
row_diff = lambda i, k: np.r_[0, np.eye(a)[i] - np.eye(a)[k], np.zeros(b)]
print("rank:", np.linalg.matrix_rank(X), " (a + b - 1 =", a + b - 1, ")")
print("alpha_1 - alpha_2 estimable:", estimable(row_diff(0, 1), X))
print("alpha_1 - alpha_3 estimable:", estimable(row_diff(0, 2), X))
X_linked = additive_matrix(cells + [(2, 2)], a, b) # one more
                                occupied cell
print("after adding cell (3, 3): rank", np.linalg.matrix_rank(X_linked),
      " alpha_1 - alpha_3 estimable:", estimable(row_diff(0, 2), X_linked))
```

[Run this code online](#)

A complete design, with every cell occupied, is connected. So is any design in which one level of B appears with every level of A and one level of A with every level of B . Disconnected designs arise in practice more often than one might expect: in observational data with many sparse levels (workers and firms, students and schools, players and teams), and in experiments where blocks were filled carelessly. The estimable functions of a disconnected design are characterized in [Exercise \(C1\)](#).

8.6.2 Two factors with interaction

The additive model assumes that the difference between two levels of A is the same at every level of B . The *interaction model* drops that assumption:

$$E(y) = \mu + \alpha_i + \beta_j + \gamma_{ij} \quad \text{for an observation in cell } (i, j), \quad (8.6.2)$$

with an interaction parameter γ_{ij} for each of the ab cells. Write $\mathbf{W}$ for the $n \times ab$ indicator matrix of the cells. The column of an empty cell is zero. The model matrix is $\mathbf{X} = [\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B, \mathbf{W}]$, with $p = 1 + a + b + ab$ columns, and m denotes the number of occupied cells.

Proposition 8.6.2. Estimability in the interaction model

In the interaction model [Eq. 8.6.2](#) with m occupied cells, write $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$ for the mean of an occupied cell.

- (a) $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{W})$ and $\text{rank}(\mathbf{X}) = m$: the model is the cell-means model in disguise.
- (b) The estimable functions are exactly the linear combinations $\sum c_{ij}\mu_{ij}$ over occupied cells.
- (c) An interaction contrast $\sum_{i,j} d_{ij}\gamma_{ij}$ whose coefficient table (d_{ij}) has zero row and column sums is estimable iff $d_{ij} = 0$ for every empty cell. In particular, $\gamma_{ij} - \gamma_{il} - \gamma_{kj} + \gamma_{kl}$ is estimable iff all four cells are occupied.

Proof.

- (a) Every column of $\mathbf{1}$, $\mathbf{Z}_A$ and $\mathbf{Z}_B$ is a sum of columns of $\mathbf{W}$. For example, the indicator of level i of A is the sum of the indicators of the cells in row i . So $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{W})$. The nonzero columns of $\mathbf{W}$ are the m indicators of occupied cells, which have disjoint supports.
- (b) By Theorem 8.2.2, the estimable functions are the combinations of the expected responses, and these are the μ_{ij} .
- (c) If $d_{ij} = 0$ off the occupied cells, then $\sum_{\text{occ}} d_{ij}\mu_{ij} = \sum d_{ij}\mu + \sum_i \alpha_i \sum_j d_{ij} + \sum_j \beta_j \sum_i d_{ij} + \sum d_{ij}\gamma_{ij} = \sum d_{ij}\gamma_{ij}$, because the row and column sums vanish. So the contrast is estimable. Conversely, if $d_{i_0j_0} \neq 0$ for an empty cell (i_0, j_0) , the column of $\gamma_{i_0j_0}$ in $\mathbf{X}$ is zero, so the coordinate vector $\mathbf{v}$ of that parameter lies in $\mathcal{N}(\mathbf{X})$. The contrast's coefficient vector has inner product $d_{i_0j_0} \neq 0$ with $\mathbf{v}$, so it is not estimable (Theorem 8.3.1(c)).

■

In the interaction model the fitted value of each occupied cell is its own mean, and an empty cell has no estimable mean at all. The two models treat empty cells in opposite ways. The additive model fills them by assumption, while the interaction model declines to say anything about them. Which is appropriate is a substantive question, and [Chapter 17](#) returns to it (Theorem 17.4.1).

Example. Empty cells in a survey

Cross-classify the 944 ANES respondents of Example (Party identification and ideology) by party identification (seven levels) and education (seven levels, labelled in the statsmodels documentation from “grades 1–8” to “PhD”). Two of the 49 cells are empty: no pure independent and no weak Republican in the sample has only an elementary-school education. Other cells are thin: only 3 pure independents report some high school. The overparameterized interaction model has $p = 64$ columns and rank 47, the number of occupied cells, as [Proposition 8.6.2\(a\)](#) says. The interaction contrast comparing strong Democrats with pure independents across “some high school” and “high-school graduate” involves four occupied cells and is estimable. It rests on only 3 respondents in one cell, so it is imprecise, but estimable. The same comparison across “grades 1–8” and “high-school graduate” involves an empty cell and is not estimable. No amount of computation will produce it, and software that prints a number for it is printing an artefact of its side condition. The additive model, with rank $7 + 7 - 1 = 13$, treats both comparisons as zero by assumption.

```
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
pid = anes["PID"].to_numpy().astype(int)           # 7 levels
educ = anes["educ"].to_numpy().astype(int) - 1     # 7 levels, now 0..6
counts = np.zeros((7, 7), dtype=int)
np.add.at(counts, (pid, educ), 1)
print("empty (PID, educ) cells:",
      [(int(i), int(j) + 1) for i, j in zip(*np.nonzero(counts == 0))])

n = len(pid)
cell = pid * 7 + educ
X_int = np.column_stack([np.ones(n), np.eye(7)[pid], np.eye(7)[educ], np.eye(49)[
    cell]])
print("interaction model: p =", X_int.shape[1], " rank =", np.linalg.matrix_rank(
    X_int),
      " occupied cells =", int((counts > 0).sum()))

def interaction_contrast(i, k, j, l):
    """gamma_ij - gamma_il - gamma_kj + gamma_kl, as a coefficient vector on X_int.
    """
    lam = np.zeros(X_int.shape[1])
    for (r, c), s in [((i, j), 1), ((i, l), -1), ((k, j), -1), ((k, l), 1)]:
        lam[15 + r * 7 + c] += s
    return lam

print("PID 0 vs 3, educ 2 vs 3:", estimable(interaction_contrast(0, 3, 1, 2), X_int)
      )
print("PID 0 vs 3, educ 1 vs 3:", estimable(interaction_contrast(0, 3, 0, 2), X_int)
      )
```

[Run this code online](#)

8.6.3 A factor and a numeric covariate

Now let a factor with g levels, with indicator matrix $\mathbf{Z}$, share the model with a numeric covariate $\mathbf{x}$. The two basic models are the **parallel lines** model

$$E(y) = \mu_k + \theta x \quad \text{for an observation at level } k, \quad (8.6.3)$$

with model matrix $[\mathbf{Z}, \mathbf{x}]$, and the **separate lines** model

$$E(y) = \mu_k + \theta_k x, \quad (8.6.4)$$

with model matrix $[\mathbf{Z}, \mathbf{Z} \circ \mathbf{x}]$, where $\mathbf{Z} \circ \mathbf{x}$ multiplies each indicator column entrywise by $\mathbf{x}$, so its k th column is $\mathbf{x}$ on level k and zero elsewhere. The same models can be written with an intercept and any coding of the factor. For example, the separate lines model in reference coding has columns $\mathbf{1}$, $\mathbf{ZC}$, $\mathbf{x}$, $(\mathbf{ZC}) \circ \mathbf{x}$. The coefficients of the last block are then differences of slopes from the reference level, and the column space is unchanged.

By Exercise (B1), the common slope θ is identifiable iff $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$, that is, iff x varies within at least one level. In the separate lines model, the slope θ_k is identifiable iff x varies within level k , since the k th column of $\mathbf{Z} \circ \mathbf{x}$ is a multiple of $\mathbf{z}_k$ exactly when x is constant on level k . When the slopes are identifiable, the separate lines model is simply g simple regressions fitted at once. Its coefficients are those of the separate fits, because the columns for different levels have disjoint supports.

The parallel lines model is where adjustment happens.

Proposition 8.6.3. Adjusted differences

In the parallel lines model Eq. 8.6.3 with $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$, let $\bar{y}_k$ and $\bar{x}_k$ be the level means. Then

$$\hat{\theta} = \frac{\sum_k \sum_{i \in k} (x_i - \bar{x}_k)(y_i - \bar{y}_k)}{\sum_k \sum_{i \in k} (x_i - \bar{x}_k)^2}, \quad \hat{\mu}_k - \hat{\mu}_l = (\bar{y}_k - \bar{y}_l) - \hat{\theta}(\bar{x}_k - \bar{x}_l).$$

Proof. The model matrix $[\mathbf{Z}, \mathbf{x}]$ has full column rank because $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$. Apply Theorem 6.6.2 with $\mathbf{X}_1 = \mathbf{x}$ and $\mathbf{X}_2 = \mathbf{Z}$. Projecting onto $\mathcal{C}(\mathbf{Z})$ replaces each entry by its level mean (Example (The one-way layout)), so $(\mathbf{I} - \mathbf{M}_Z)\mathbf{x}$ and $(\mathbf{I} - \mathbf{M}_Z)\mathbf{y}$ are the within-level deviations, and $\hat{\theta}$ is the no-intercept slope of one on the other. This is the first formula. By Theorem 6.6.2(c), $\hat{\boldsymbol{\mu}} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top (\mathbf{y} - \hat{\theta} \mathbf{x})$, whose k th entry is $\bar{y}_k - \hat{\theta} \bar{x}_k$. Subtracting gives the second formula. ■

The difference $\bar{y}_k - \bar{y}_l$ compares the levels as they happen to be. The adjusted difference compares them at equal values of the covariate, correcting for the fact that the groups differ in x . The correction is the slope times the difference in covariate means. It is large when the groups are unbalanced on a covariate that matters. The estimated slope uses only variation *within* levels, so that differences between levels in both x and y cannot be mistaken for an effect of x .

Example. Education and television news, adjusted for age

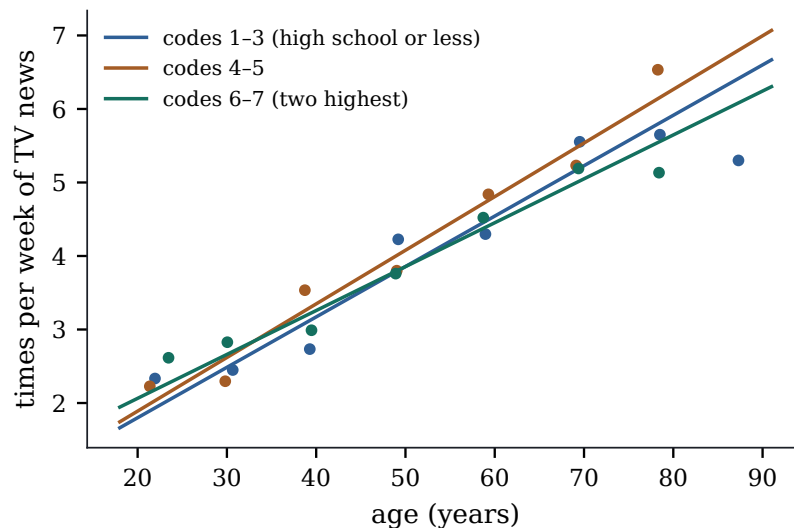


Figure 8.6.1: Times per week of television news against age for three education groups in the 1996 ANES: separate least squares lines (Eq. 8.6.4) and the group means within ten-year age bins (points).

In the ANES data, respondents report how many times per week they watch television news. Coding education with seven levels and “high-school graduate” as the reference, respondents with some high school but no diploma watch 1.005 times a week more than high-school graduates. They are also older, with mean ages 59.8 and 48.2, and news watching rises with age. Adding age to the model, the within-level slope is 0.0674 times per week per year of age, and the adjusted difference shrinks to 0.224. More than three quarters of the raw difference is an age difference, and Proposition 8.6.3 accounts for it exactly. Whether the adjusted figure is the “effect of education” is a different question. Here age plausibly influences both schooling (older cohorts had less of it) and viewing habits, which argues for adjusting, but Chapter 25 shows that adjustment is not always appropriate.

Allowing separate age slopes for three education groups (codes 1–3, high school or less, 313 respondents; codes 4–5, 277; and the two highest levels, codes 6–7, 354) gives slopes 0.0686, 0.0729 and 0.0597, close to the common slope 0.0672 of the parallel lines model with the same three groups. The residual sum of squares falls only from 5623.6 to 5616.7 for the two extra parameters. Figure 8.6.1 shows the three fitted lines with binned means. The lines are nearly parallel, and the highest education group starts higher and rises a little more slowly.

```
tv = anes["TVnews"].to_numpy()                                # times per week watching TV
news
age = anes["age"].to_numpy()
E = np.eye(7)[educ][:, [0, 1, 3, 4, 5, 6]]                  # reference: high-school
graduate
unadjusted = np.linalg.lstsq(np.column_stack([np.ones(n), E]), tv, rcond=None)[0]
adjusted = np.linalg.lstsq(np.column_stack([np.ones(n), E, age]), tv, rcond=None)[0]
print("some high school minus high-school graduate:",
      f"unadjusted {unadjusted[2]:.3f}, adjusted for age {adjusted[2]:.3f}")
print("mean age: some high school", age[educ == 1].mean().round(1),
      " high-school graduate", age[educ == 2].mean().round(1),
      " common age slope", adjusted[-1].round(4))
```

[Run this code online](#)

```

grp = np.select([educ <= 2, educ <= 4], [0, 1], 2) # education codes 1-3 / 4-5 /
6-7
D = np.eye(3)[grp]
X_par = np.column_stack([D, age]) # parallel lines: 3 intercepts,
one slope
X_sep = np.column_stack([D, D * age[:, None]]) # separate lines: 3 intercepts,
3 slopes
b_par = np.linalg.lstsq(X_par, tv, rcond=None)[0]
b_sep = np.linalg.lstsq(X_sep, tv, rcond=None)[0]
sse_par = np.sum((tv - X_par @ b_par) ** 2)
sse_sep = np.sum((tv - X_sep @ b_sep) ** 2)
print("separate slopes:", b_sep[3:].round(4),
      f" SSE parallel {sse_par:.1f}, separate {sse_sep:.1f}")

```

[Run this code online](#)

Warning. In a separate lines model the difference between two levels is $(\mu_k - \mu_l) + (\theta_k - \theta_l)x$, which depends on x . The coefficient that software labels as the “main effect” of level k is this difference at $x = 0$. For a covariate like age, that is a comparison of newborns, far outside the data. Centre the covariate at a meaningful value before reading such coefficients, or report the difference at several values of x .

8.6.4 Broken lines

Indicators can also switch a slope on. Let c be a known value of x and put $(x-c)_+ = \max(x-c, 0)$, which is $x - c$ times the indicator of $x > c$. The model

$$E(y) = \beta_0 + \beta_1 x + \beta_2 (x - c)_+$$

is a continuous line with a bend at c : slope β_1 to the left of c and $\beta_1 + \beta_2$ to the right. Adding the indicator $\mathbf{1}\{x > c\}$ itself allows a jump at c as well. With the bend point known, these are ordinary linear models, and all of this chapter applies. For example, β_2 is identifiable iff the data contain values of x on both sides of c (strictly below and strictly above) and at least three distinct values of x in all (Exercise (B2)). When c is unknown the model is nonlinear in c . Such piecewise linear terms are the simplest splines (Chapter 42, Definition 42.3.1), the subject of Part IX.

8.6.5 Exercises

8.6.5.1 A. Check your understanding

Exercise. A1

In the additive model with $a = b = 2$, suppose cells $(1, 1)$, $(2, 1)$ and $(2, 2)$ are occupied and $(1, 2)$ is empty. Find the rank of $\mathbf{X}$ and a basis of $\mathcal{N}(\mathbf{X})$. Show that $\mu + \alpha_1 + \beta_2$, the mean of the empty cell, is estimable by writing it as a combination of the means of occupied cells.

Exercise. A2

A factor has three levels, and the covariate x is constant on level 2 but varies on levels 1 and 3. Find the rank of the separate lines model matrix $[\mathbf{Z}, \mathbf{Z} \circ \mathbf{x}]$, and say which of

$\mu_1, \mu_2, \mu_3, \theta_1, \theta_2, \theta_3$ are identifiable.

8.6.5.2 B. Practice

Exercise. B1

In the separate lines model, show that the mean at level k and covariate value x_0 , $\mu_k + \theta_k x_0$, is estimable iff either x varies within level k , or x_0 equals the common value of x on level k . Interpret this as a statement about extrapolation.

Solution. The mean is $\lambda^\top \beta$ with λ having 1 in the position of μ_k and x_0 in the position of θ_k . The columns for μ_k and θ_k are $\mathbf{z}_k$ and $\mathbf{z}_k \circ \mathbf{x}$, and they involve only level k . If x varies on level k , both coefficients are identifiable (Proposition 8.1.3), so $\lambda^\top \beta$ is estimable. If $x \equiv x^*$ on level k , then $\mathbf{z}_k \circ \mathbf{x} = x^* \mathbf{z}_k$, and the null space contains the vector with $-x^*$ at μ_k and 1 at θ_k . Then $\lambda^\top \mathbf{v} = x_0 - x^*$, which vanishes iff $x_0 = x^*$. A single covariate value within a level supports no statement about other values, however many observations share it.

Exercise. B2

Write the broken-line model $E(y) = \beta_0 + \beta_1 x + \beta_2 (x - c)_+$ in matrix form and prove the identifiability condition stated in the text. Show that the continuous broken line is the same as the model with two separate lines, one on each side of c , constrained to meet at c , and count parameters to explain the difference of one.

Exercise. B3

In the interaction model, suppose all cells are occupied except $(1, 1)$. Show that the $(a - 1)(b - 1)$ contrasts $\gamma_{ij} - \gamma_{i1} - \gamma_{1j} + \gamma_{11}$ are no longer estimable, but that $\gamma_{ij} - \gamma_{i2} - \gamma_{1j} + \gamma_{12}$ is estimable for $i \geq 2, j \geq 3$. How many linearly independent estimable interaction contrasts remain?

8.6.5.3 C. Going deeper

Exercise. C1

In the additive model, let the design graph have components $C_1, \dots, C_c$. Show that $\lambda_0 \mu + \sum_i \lambda_i \alpha_i + \sum_j \kappa_j \beta_j$ is estimable iff $\lambda_0 = \sum_j \kappa_j$ and, for every component C , $\sum_{i \in C} \lambda_i = \sum_{j \in C} \kappa_j$. Deduce Theorem 8.6.1(b).

Solution. From the proof of Theorem 8.6.1, the null vectors are $(m, \mathbf{u}, \mathbf{v})$ with $u_i = t_{C(i)}$ and $v_j = -m - t_{C(j)}$, for arbitrary $m, t_1, \dots, t_c$. The inner product with the coefficient vector is

$$\lambda_0 m + \sum_i \lambda_i t_{C(i)} - \sum_j \kappa_j (m + t_{C(j)}) = m \left(\lambda_0 - \sum_j \kappa_j \right) + \sum_C t_C \left(\sum_{i \in C} \lambda_i - \sum_{j \in C} \kappa_j \right).$$

This vanishes for all m, t iff every bracket is zero, which is the stated condition (Theo-

rem 8.3.1(c)). For $\alpha_i - \alpha_k$, $\lambda_0 = 0$, all $\kappa_j = 0$, and the component sums of λ are zero iff i and k lie in the same component.